

AI-tool adoption and self-reported job satisfaction among professional software developers: a descriptive analysis of the Stack Overflow Annual Developer Survey

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Abstract

This is a descriptive study. Using the Stack Overflow Annual Developer Survey 2025, we document the cross-sectional relationship between professional developers’ reported adoption of AI coding tools and their reported job satisfaction, controlling for years of professional experience and primary developer role. In an analysis sample of 24,318 professional developers with complete data, daily and occasional AI-tool users report job satisfaction about 0.15 points higher (on a 0–10 scale) than non-users after adjustment, a difference that is statistically distinguishable from zero ($p < 0.001$) but small in magnitude: the specification explains only about 2% of the variation in satisfaction. We make no causal claim. The survey is observational, selection into it is not random, and both AI-tool adoption and satisfaction are self-reported, with no instrument for adoption and no longitudinal variation. We spell out these limits in detail. This paper is a teaching artefact produced for a seminar on AI for research; it is not intended for publication.

1 Introduction

AI coding assistants moved, in a short span, from novelty to default part of many developers’ daily workflow. A natural and frequently asked question is whether developers who adopt these tools are, on average, more satisfied with their jobs — whether because the tools remove drudgery, or simply because the kind of developer who adopts them differs in ways that also relate to satisfaction. The Stack Overflow Annual Developer Survey, which reaches tens of thousands of developers and asks about both AI-tool use and job satisfaction, is a convenient place to look at this association descriptively.

We emphasise at the outset that this is a descriptive exercise. We document a correlation in a single cross-section of self-reported data; we do not estimate the effect of AI-tool adoption on satisfaction, and nothing in our design would license that interpretation. AI-tool adoption is a choice that is plausibly correlated with developer ability, team culture, employer, and even with satisfaction itself. We therefore report associations, flag the threats to any causal reading prominently (Section 5), and resist causal language throughout.

Our contribution is pedagogical rather than substantive. The paper is a transparent, end-to-end worked example for a seminar on using AI tools in the research workflow: a public dataset, a clearly stated descriptive question, a small reproducible analysis, and an honest accounting of what the data can and cannot support. The “finding” is incidental; the worked example is the point.

2 Data

We use the Stack Overflow Annual Developer Survey 2025 [Stack Overflow, 2025], a large annual online survey of software developers recruited primarily through Stack Overflow’s own channels. Participation is voluntary and unincentivised, so the sample is one of convenience rather than a probability sample of any defined developer population. The public release contains 49,191 responses. The data are distributed under the Open Database License (ODbL) 1.0 and the Database Contents License (DbCL) 1.0; we attribute Stack Overflow as the source as those licenses require.

We restrict attention to respondents who identify as professional developers (`MainBranch = "I am a developer by profession"`), of whom there are 37,467. Our key variables are: reported AI-tool use (`AISelect`); job satisfaction (`JobSat`), a self-rating on a 0–10 scale; years of professional work experience (`WorkExp`); and primary developer role (`DevType`). We note that the 2025 instrument records general professional work experience rather than a dedicated “years coding professionally” field; we use `WorkExp` as the experience control and flag this as a measurement choice rather than an exact match to the construct.

We construct a binary AI-use indicator equal to one for respondents who report using AI tools daily, weekly, or monthly/infrequently, and zero for those who report not using them (whether or not they plan to). Dropping respondents with missing values on any of the four variables, and retaining developer-role categories with at least 30 observations for stable fixed effects, leaves an analysis sample of **24,318** professional developers across 30 role categories. Missingness is substantial — job satisfaction in particular is unanswered by a large share of respondents — and is almost certainly not random; we return to this in Section 5. Figure 2 shows the distribution of professional experience in the analysis sample, which is right-skewed with a median of 11 years.

3 Method

We estimate by ordinary least squares the cross-sectional association

$$\text{JobSat}_i = \beta_0 + \beta_1 \text{AIUser}_i + \beta_2 \text{WorkExp}_i + \gamma_{\text{role}(i)} + \varepsilon_i,$$

where AIUser_i is the binary AI-tool-use indicator, WorkExp_i is years of professional experience, and $\gamma_{\text{role}(i)}$ are developer-role fixed effects. The coefficient of interest is β_1 , the adjusted mean difference in reported satisfaction between AI-tool users and non-users.

We stress that β_1 is a conditional correlation, not an effect. Because AI-tool adoption is not randomly assigned and is plausibly related to unobserved determinants of satisfaction, the regression removes confounding only along the dimensions we observe and include (experience and role). We report it as a descriptive summary of the data, and we assess its stability across alternative specifications rather than treating any one number as an estimate of a causal parameter.

4 Results

Figure 1 shows mean job satisfaction by reported AI-tool use. The four non-daily categories are nearly indistinguishable, clustered around 7.1, while daily users sit slightly higher at 7.30. Collapsed to the binary indicator, AI-tool users report mean satisfaction of 7.24 versus 7.13 for non-users, a raw gap of about 0.10 points on the 0–10 scale.

Table 1 reports the regressions. Unadjusted, the AI-user indicator carries a coefficient of 0.104 ($p = 0.001$). Adding the experience control and role fixed effects (column 2, our main specification) the coefficient is 0.151 (standard error 0.033; $t = 4.63$; $p < 0.001$; 95% CI [0.087, 0.215]). The point estimate is precisely determined — a consequence of the large sample — but small:

a sixth of a point on a ten-point scale. The model’s R^2 is only 0.021, so AI-tool use, experience, and role together account for about 2% of the variation in reported satisfaction. Experience itself enters positively and strongly, at 0.020 satisfaction points per additional year ($p < 0.001$).

The sign and significance of the AI-use coefficient are stable across alternatives. Replacing the binary indicator with an ordinal use-frequency score (column 3) yields 0.059 per step up the frequency ladder ($p < 0.001$). Restricting the sample to developers with more than five years of professional experience (column 4) leaves the association essentially unchanged at 0.186 ($p < 0.001$, $n = 18,313$). We therefore document a *small, robust, positive cross-sectional association* between reported AI-tool adoption and reported job satisfaction; controlling for experience and role does not change its sign or significance, and indeed slightly increases it.

Table 1: OLS regressions of job satisfaction (0–10) on AI-tool use. Column 2 is the main specification. Standard errors in parentheses.

	(1) Unadj.	(2) Main	(3) Ordinal	(4) >5y exp.
AI user (binary)	0.104 (0.032)	0.151 (0.033)		0.186 (0.036)
AI-use frequency (0–4)			0.059 (0.009)	
Years experience		0.020 (0.001)	0.020 (0.001)	0.018 (0.001)
Role fixed effects	No	Yes	Yes	Yes
N	24,318	24,318	24,318	18,313
R^2	0.000	0.021	0.022	0.018

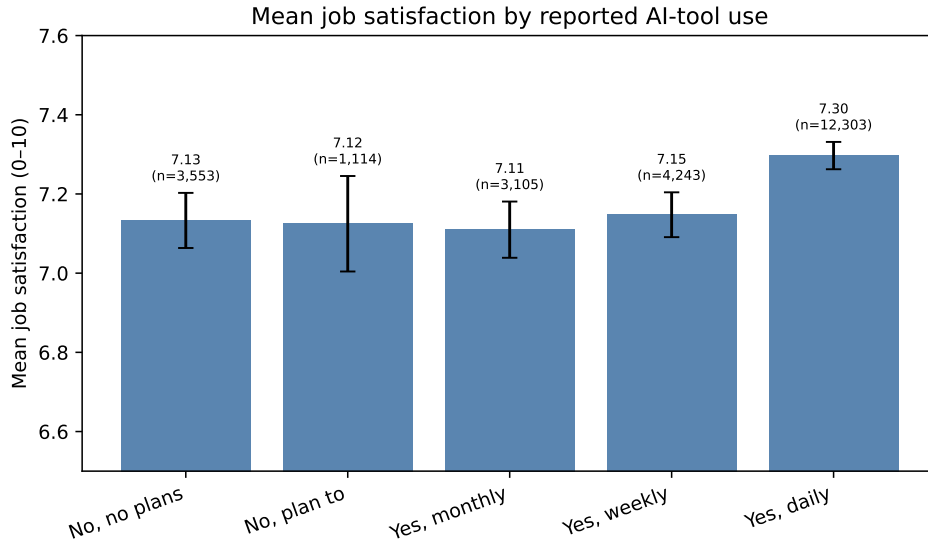


Figure 1: Mean self-reported job satisfaction (0–10) by reported AI-tool use, among 24,318 professional developers in the analysis sample. Bars show category means; whiskers are 95% confidence intervals; cell sizes are printed above each bar. The four non-daily categories cluster near 7.1; daily users are slightly higher at 7.30.



Figure 2: Distribution of years of professional work experience among the 24,318 professional developers in the analysis sample. The distribution is right-skewed with a median of 11 years (dashed line); the long right tail reflects a minority of very experienced respondents.

5 Limitations

The limitations are the most important part of this paper, because they, not the coefficient, determine what may be concluded. The headline point is simple: *nothing here supports a causal interpretation*.

Selection into the survey is not random. The Stack Overflow survey is a voluntary convenience sample recruited largely through Stack Overflow’s own audience. Developers who take the survey differ from those who do not, and the differences plausibly relate to both AI-tool use and job satisfaction. The sample is not representative of any defined population of developers, so even the descriptive means generalise only with caution.

AI-tool adoption is self-reported. We observe what respondents say about their AI-tool use, not their actual use. Self-reports may be coloured by enthusiasm, social desirability, or differing interpretations of what counts as “using AI tools.”

Job satisfaction is self-reported. The outcome is a single 0–10 self-rating collected at one moment. It is subjective, sensitive to mood and framing, and not anchored to any objective benchmark. Common self-report method bias could inflate the association if the same disposition pushes a respondent to report both higher tool use and higher satisfaction.

There is no instrument for AI-tool adoption. We have no source of variation in adoption that is plausibly unrelated to satisfaction except through adoption. Without such an instrument (or an experiment), we cannot separate any effect of the tools from selection into using them.

The design is cross-sectional. All variables are measured at once. We cannot observe a developer’s satisfaction before and after adopting AI tools, so we cannot rule out reverse causation: more satisfied developers may simply be more inclined to try new tools.

Omitted variables are plausibly correlated with both sides. Developer ability, team culture, employer quality, compensation, and autonomy are all things we do not adequately observe, and each could raise both AI-tool adoption and job satisfaction. Our controls — experience and role — absorb only a fraction of this, consistent with the model’s very low R^2 . The adjusted coefficient should be read as “the satisfaction gap that remains after accounting for experience and role,” not as an effect net of confounding.

Finally, missingness is heavy and non-random: many respondents do not answer the satisfaction item at all, and those who skip it may differ systematically from those who answer. Our complete-case analysis ignores this and could be biased as a result.

6 Conclusion

We document a small, statistically robust, positive cross-sectional association between reported AI-tool adoption and reported job satisfaction among professional developers in the Stack Overflow 2025 survey: roughly 0.15 points on a 0–10 scale after adjusting for experience and role, stable across specifications but explaining only about 2% of the variation in satisfaction. We did *not* find, and did not attempt to estimate, an effect of AI tools on satisfaction; the association is consistent with selection, reverse causation, and omitted variables in equal measure.

This paper is a teaching artefact for a seminar on AI for research. Its purpose is to show the full loop — from a public dataset to a short, reproducible, honestly hedged write-up — and to make visible where such analyses are weakest: not in the arithmetic, which is easy, but in the identification and the citations, which are not. Readers should take away the workflow and the discipline of the limitations section, not the coefficient.

References

Stack Overflow. Annual developer survey 2025. <https://survey.stackoverflow.co/>, 2025.
Licensed under ODbL 1.0 and DbCL 1.0. Replace year if a different snapshot is used.